

Creating the Repository

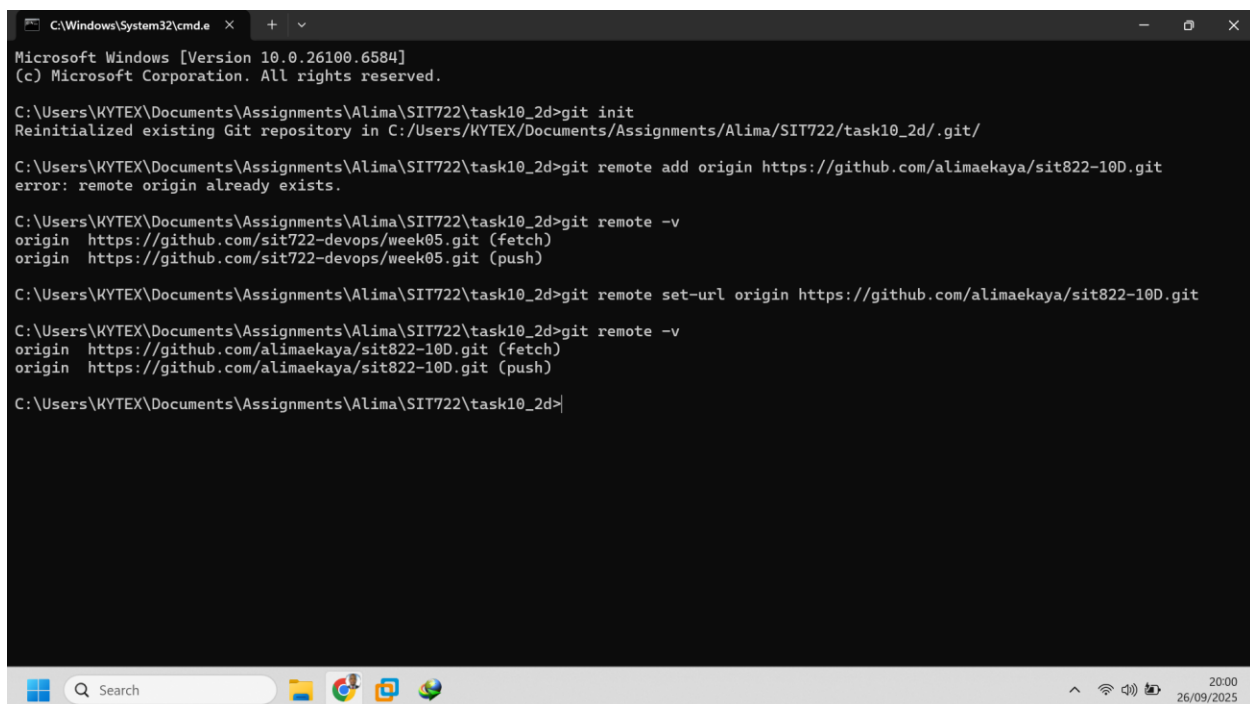
I began by creating a new empty GitHub repository named *alimaekaya-sit722* as instructed. I followed the official GitHub guide on how to create a new repository, ensuring that it was initialized without a README, license, or .gitignore. This prepared a clean workspace to connect with my local project later.

Setting up the Local Repository

On my local machine, I created a new project folder and used the command line instructions provided by GitHub after repository creation. These included initializing the folder as a Git repository, adding the remote origin, and confirming the connection. This ensured that my local project and the remote repository were linked correctly.

Downloading and Uploading Project Code

I then downloaded the provided Task 10.2D resource code, unzipped it, and placed the files into my local project folder. Once the code was in place, I added all files to Git, committed them with a clear message, and pushed the initial version to the remote repository on GitHub. At this point, the project code was version-controlled and available online.



```
C:\Windows\System32\cmd.exe X + v
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git init
Reinitialized existing Git repository in C:/Users/KYTEX/Documents/Assignments/Alima/SIT722/task10_2d/.git/

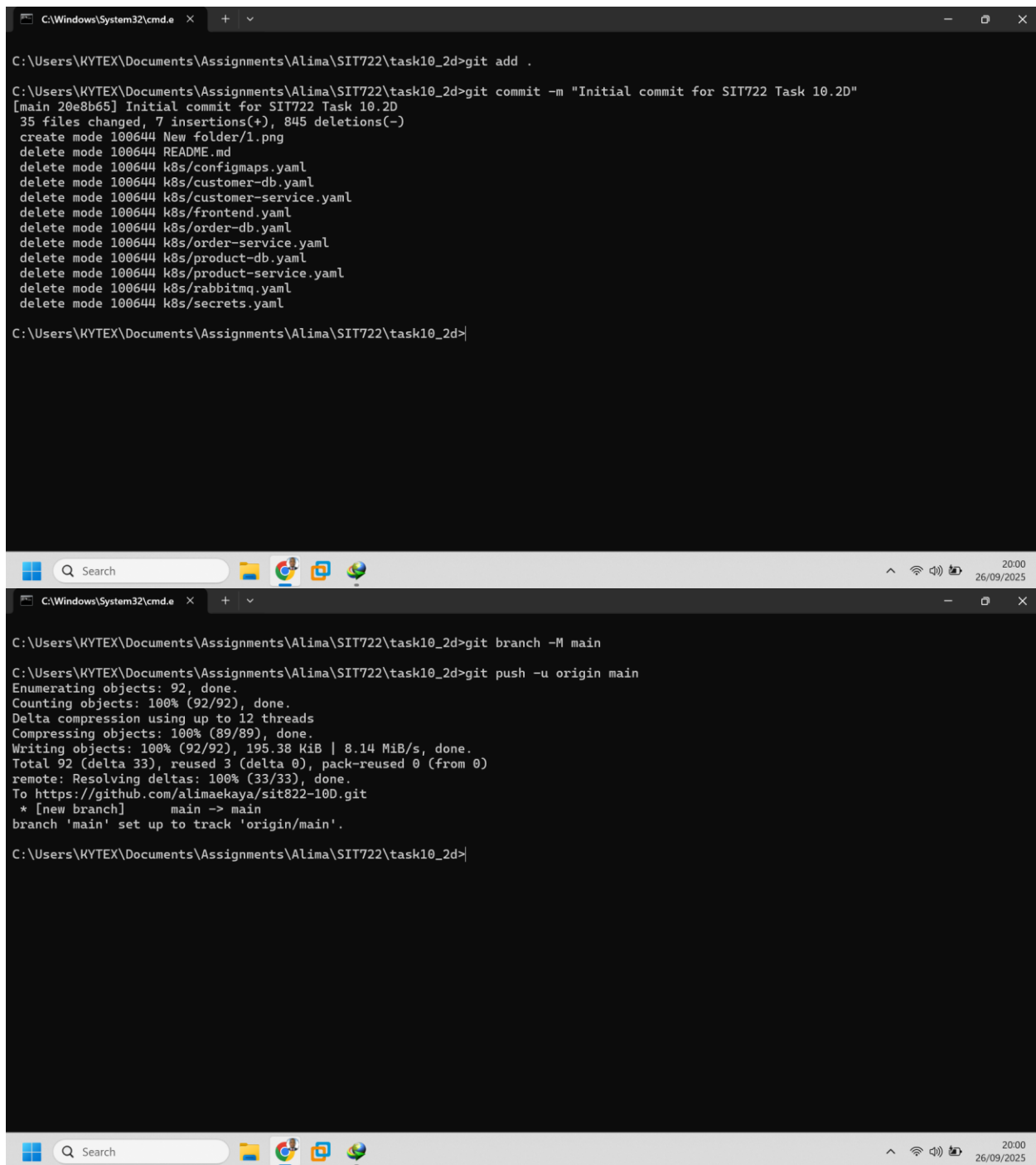
C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git remote add origin https://github.com/alimaekaya/sit822-10D.git
error: remote origin already exists.

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git remote -v
origin https://github.com/sit722-devops/week05.git (fetch)
origin https://github.com/sit722-devops/week05.git (push)

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git remote set-url origin https://github.com/alimaekaya/sit822-10D.git

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git remote -v
origin https://github.com/alimaekaya/sit822-10D.git (fetch)
origin https://github.com/alimaekaya/sit822-10D.git (push)

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>
```



```
C:\Windows\System32\cmd.exe X + v

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git add .

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git commit -m "Initial commit for SIT722 Task 10.2D"
[main 20e8b65] Initial commit for SIT722 Task 10.2D
35 files changed, 7 insertions(+), 845 deletions(-)
create mode 100644 New folder/1.png
delete mode 100644 README.md
delete mode 100644 k8s/configmaps.yaml
delete mode 100644 k8s/customer-db.yaml
delete mode 100644 k8s/customer-service.yaml
delete mode 100644 k8s/frontend.yaml
delete mode 100644 k8s/order-db.yaml
delete mode 100644 k8s/order-service.yaml
delete mode 100644 k8s/product-db.yaml
delete mode 100644 k8s/product-service.yaml
delete mode 100644 k8s/rabbitmq.yaml
delete mode 100644 k8s/secrets.yaml

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>

C:\Windows\System32\cmd.exe X + v

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git branch -M main

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git push -u origin main
Enumerating objects: 92, done.
Counting objects: 100% (92/92), done.
Delta compression using up to 12 threads
Compressing objects: 100% (89/89), done.
Writing objects: 100% (92/92), 195.38 KiB | 8.14 MiB/s, done.
Total 92 (delta 33), reused 3 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (33/33), done.
To https://github.com/alimaekaya/sit822-10D.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>
```

Creating the CI Pipeline (Stage 1)

Next, I designed and implemented a complete CI pipeline using GitHub Actions. The pipeline was configured to trigger automatically whenever code was pushed to the testing branch. Its main purpose was to run build steps and execute tests on the code. Only if all tests passed successfully would the pipeline proceed to build a Docker image and upload it to DockerHub. This setup ensured that only tested and verified code could move forward.

```
C:\Windows\System32\cmd.exe X + v

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git branch -M main

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git push -u origin main
Enumerating objects: 92, done.
Counting objects: 100% (92/92), done.
Delta compression using up to 12 threads
Compressing objects: 100% (89/89), done.
Writing objects: 100% (92/92), 195.38 KiB | 8.14 MiB/s, done.
Total 92 (delta 33), reused 3 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (33/33), done.
To https://github.com/alimaekaya/sit822-10D.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git checkout -b testing
Switched to a new branch 'testing'

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git push -u origin testing
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'testing' on GitHub by visiting:
remote:   https://github.com/alimaekaya/sit822-10D/pull/new/testing
remote:
To https://github.com/alimaekaya/sit822-10D.git
 * [new branch]      testing -> testing
branch 'testing' set up to track 'origin/testing'.

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>
```

Staging Deployment (Stage 2)

Once the image was uploaded, the pipeline was extended to create a temporary staging environment. Using workflow actions, the Docker image was deployed in this staging environment, allowing me to run manual or basic acceptance tests. After confirming that the system worked as expected, the staging environment was destroyed to save resources. This step acted as a controlled checkpoint before production deployment.

```
C:\Windows\System32\cmd.exe X + v
C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git checkout -b testing
fatal: a branch named 'testing' already exists

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>git push -u origin testing
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 860 bytes | 430.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/alimaekaya/sit822-10D.git
  20e8b65..ae802cb  testing -> testing
branch 'testing' set up to track 'origin/testing'.

C:\Users\KYTEX\Documents\Assignments\Alima\SIT722\task10_2d>|
```

Production Deployment

Finally, I merged the tested code from the testing branch into the main branch through a pull request. This merge action triggered the production deployment pipeline, which deployed the latest stable Docker image to the existing production environment. This ensured that only thoroughly tested and approved code reached production, supporting a reliable and secure DevOps workflow.

Actions secrets - alimaekaya/sit822-10D

login.docker.com/u/login/ident... | Google Password Manager

github.com/alimaekaya/sit822-10D/settings/secrets/actions

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models

Webhooks

Copilot

Environments

Codespaces

Pages

Security

Advanced Security

Deploy keys

Secrets and variables

Actions

Codespaces

Dependabot

Integrations

GitHub Apps

Email notifications

Secrets and variables allow you to manage reusable configuration data. Secrets are encrypted and are used for sensitive data. Variables are shown as plain text and are used for non-sensitive data.

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets Variables

Environment secrets

This environment has no secrets.

Manage environment secrets

Repository secrets

New repository secret

Name	Last updated
ALIMAEEKAYA	now

Search

Fix pipeline: build & push front... | login.docker.com/u/login/ident... |

github.com/alimaekaya/sit822-10D/actions/runs/18054737563

alimaekaya / sit822-10D

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

CI/CD Pipeline

Fix pipeline: build & push frontend Docker image with Nginx #4

Re-run all jobs

Summary

Jobs

build-and-push

Run details

Usage

Workflow file

Triggered via push 1 minute ago

alimaekaya pushed -> 008f6b2 testing

Status Success

Total duration 23s

Artifacts -

ci-cd.yml

on: push

build-and-push 19s

Search

20:00 26/09/2025